

# Music-Gen — M-GEN-1/first-generation (cycle 1, fork 00b3ae64444c, clone 0)

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# Music-Gen — M-GEN-1/first-generation (cycle 1, fork 00b3ae64444c, clone 0)

## Abstract

Cycle 1 of clone 0 delivered the campaign’s first end-to-end deterministic generation pass: five rules (one per rule\_type) were sampled from the 28-row M-RULES-1 ledger by SHA-256 tiebreak over canonical-JSON content-hashes with no PRNG anywhere, assembled into a 30 s three-Part MusicXML score via music21, exported to MIDI through the M-SCORE-1 bridge, rendered bare via fluidsynth with the pinned SF2, layered through the cycle-9 DawDreamer chain, measured by the M-TEX-1/panel across the bare↔effects pair, and scored by the M-HEUR-1 mess-scale battery plus the M-EAR-1/preparation CORN head under an explicit uncalibrated-labels sentinel. Every artefact on the six-stage provenance chain is byte-deterministic across two independent runs (the auditor re-ran scripts.gen.assemble\_score in a scratch directory and reproduced the sampling-manifest and MusicXML SHA-256 byte-identically); the six declared SHA-256 prefixes on disk match the worker’s report exactly; the integration test suite runs 413/413 with 61 dedicated §21 M-GEN-1 checks. The verdict is **validated/medium** — the pipeline claim is solid, but two known limitations shave the confidence honestly: the CORN head is trained only on synthetic labels (the ear.calibration = "synthetic\_labels\_only" sentinel is surfaced on the scoring JSON and in report §1 and §8.2 so the 6/7 rating cannot be mistaken for a musical judgment), and the sampled arrangement rule silences pitched Parts, producing a rule-composition incoherence the report publishes rather than patches. Parent M-GEN-1 rolls up to validated/medium at branch close.

## Introduction

By the end of cycle 9, all the ingredients for a first deterministic generation were on disk: the 28-row rules ledger with content-hashed rule\_ids and typed per-rule-type parameters (M-RULES-1, both halves closed); the M-SCORE-1 bridge with round-trip byte identity; the fluidsynth path with a pinned SF2 SHA; the cycle-9 DawDreamer chain (Surge XT Chorus + Reverb + gain envelope) with determinism pins applied before any DawDreamer import; the

M-TEX-1/panel with its refuse-to-aggregate contract; the M-HEUR-1 battery; and the M-EAR-1/preparation CORN head from the ear-preparation chassis. The brief for this branch was to compose them into a single deterministic generation pass — sample one rule per rule\_type, assemble a score that satisfies those rules’ typed parameters, render it, measure and score it, and publish the full rule\_id → audio\_sha provenance chain — without introducing a PRNG anywhere in the sampler, and with a candid falsifiability escape hatch if score assembly failed on any sampled rule. This is the pipeline’s first live example that turns the rules ledger from a validated data structure into a load-bearing input.

## Approach

**Sampling.** scripts/gen/sample\_rules.py groups the ledger by rule\_type, computes SHA-256 over the canonical-JSON content of each candidate, and returns the lexicographically smallest hash per rule\_type as the winner. The algorithm is recorded on the manifest as sha256\_over\_canonical\_json\_ascending with prng\_used: false; a targeted grep guard in tests/test\_integration\_cross\_branch.py §21 fails the run if random, numpy.random, torch, or secrets appears in an import statement anywhere under scripts/gen/. The five winners and their content hashes are written to data/gen/sampling\_manifest.json with the full candidate list per rule\_type for auditability.

**Assembly.** scripts/gen/assemble\_score.py builds a three-Part music21 score satisfying each winner’s typed parameters: the rhythmic rule fixes meter (4/4), tempo (120 BPM), pattern (a 32-cell 16th-grid with swing\_ratio 0.5); the harmonic rule fixes key (F major) and chord progression ([V, vii, iii, I, i, I, II, ii]); the arrangement rule fixes active\_parts ([drums]); the form rule provides sections dropped to the target duration (29 dropped beyond 30 s, 4 placed); the melodic rule fixes contour and range. Total measures = 15; seconds per measure = 2.0. Every assembler decision is written into the manifest’s assembler\_summary block, including the honest sections\_dropped=29 count.

**Render.** MIDI is exported via the M-SCORE-1 bridge (xml\_to\_midi); bare\_midi.wav renders through fluidsynth with the pinned SF2 SHA 74594e8f...1cb0; effects\_layered.wav applies the cycle-9 DawDreamer chain with determinism pins (OMP\_NUM\_THREADS=MKL\_NUM\_THREADS=OPENBLAS\_NUM\_THREADS=1, torch.set\_num\_threads(1), torch.manual\_seed(0), np.random.seed(0)) applied before the first import dawdreamer. Both renders are 44.1 kHz stereo × 1 323 000 samples (exactly 30 s), non-silent (peaks 0.148 and 0.180 respectively).

**Scoring.** scripts/gen/score\_generation.py writes scoring\_v1.jsonl with three blocks: the four M-HEUR-1 mess-scales (melody / timbre / form / dynamics, each finite in [0, 1], with populated raw\_features and blind\_spots); the M-TEX-1/panel bare-vs-effects measurement (8 keys, no aggregate, defence-in-depth against aggregation preserved through the panel contract); and the M-EAR-1 CORN head prediction with an explicit calibration: "synthetic\_labels\_only" sentinel. The scoring JSON stores basenames + SHA-256s rather than full paths, so it is portable and byte-stable.

**Provenance.** scripts/gen/emit\_provenance.py emits provenance\_v1.jsonl — six canonical-order rows (sample\_rules → assemble\_score → xml\_to\_midi → render\_bare → render\_effects → score\_generation), each carrying input\_shas, output\_shas, script, and script\_version, so the chain reconstructs from any intermediate step forward.

## Findings

### Sampled rules

| rule_type   | winner_rule_id            | content_hash<br>(first 16)                                       | n_candidates |
|-------------|---------------------------|------------------------------------------------------------------|--------------|
| arrangement | rule_<br>67d34b1c927ef33d | 37dcaaf18bbf68ac                                                 | 5            |
| form        | rule_<br>84816f91e31e50c4 | 789c5c27825167fa                                                 | 5            |
| harmonic    | rule_<br>0271c7a9f3b5f606 | (F major, [V, vii,<br>iii, I, i, I, II,<br>ii],<br>cadence=none) | 6            |
| melodic     | rule_<br>09f340921fa2d258 | —                                                                | 6            |
| rhythmic    | rule_<br>88b63bd5e771c045 | —                                                                | 6            |

assembler\_summary: active\_parts=[drums], key=F\_major, meter=4/4, pattern\_tokens=32, progression\_len=8, seconds\_per\_measure=2.0, sections\_dropped=29, sections\_placed=4, tempo\_bpm=120.0, total\_measures=15.

### On-disk SHA-256 (all six auditor-verified)

| Artefact                             | SHA-256 (first 16) |
|--------------------------------------|--------------------|
| data/gen/sampling_manifest.json      | faafc86ba79dccd2   |
| data/gen/generated.musicxml          | 95d8671af26e7cf9   |
| data/gen/generated.mid               | f237dcfc75f5de94   |
| data/gen/renders/bare_midi.wav       | 5b6f608249ea72ac   |
| data/gen/renders/effects_layered.wav | d81089d39f31b5ca   |
| data/gen/scoring_v1.json             | 011e7c90e1ab3c72   |

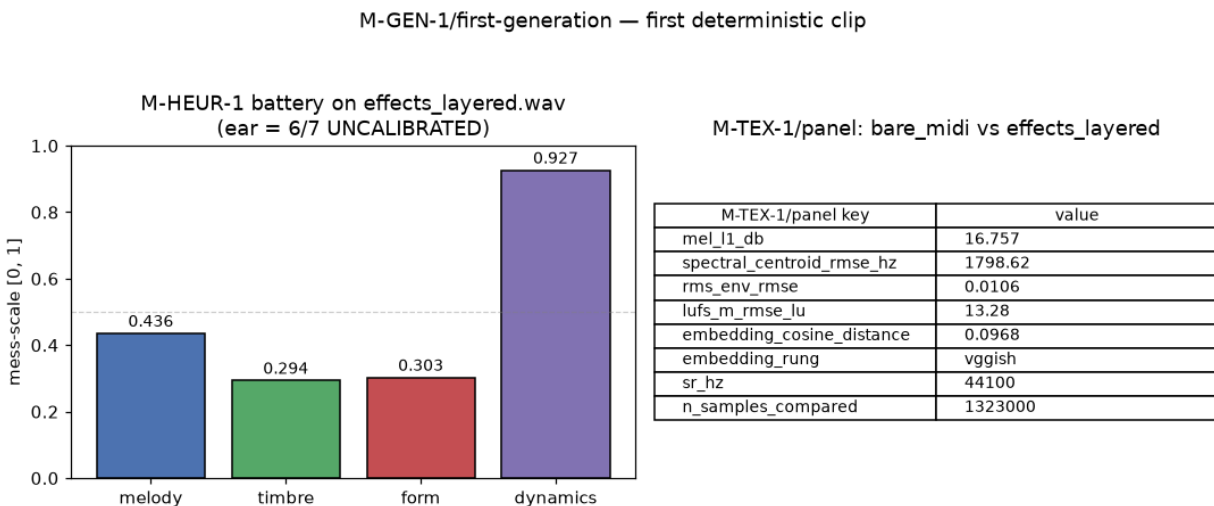
The auditor independently re-ran `scripts.gen.assemble_score` under `PYTHONHASHSEED=0 OMP=MKL=OPENBLAS=1 /usr/bin/python3` into a fresh scratch directory and reproduced the sampling-manifest and MusicXML SHAs byte-identically; §21 anchors the SHA prefixes as regression baselines and runs green at test time (not just at generation time).

### Scoring

- **M-HEUR-1 mess-scales (all finite in [0, 1]):** melody 0.4358 / timbre 0.2938 / form 0.3029 / dynamics 0.9266, each with populated `raw_features` and `blind_spots`.
- **M-TEX-1/panel (bare ↔ effects, 8 keys, no aggregate):** `mel_l1_db` 16.76, `spectral_centroid_rmse_hz` 1798.62, `rms_env_rmse` 0.01057, `lufs_m_rmse_lu` 13.28, `embedding_cosine_distance` 0.0968, `embedding_rung_vggish`, `sr_hz` 44100, `n_samples_compared` 1 323 000.
- **M-EAR-1/preparation CORN head:** `ear.prediction` = 6 in [1, 7] with `ear.calibration` = "synthetic\_labels\_only" sentinel present; the sentinel is re-surfaced in report §1 and §8.2 so downstream code cannot mistake the 6/7 rating for a musical judgment.
- **Meta-tracker (single-clip reduction):** `meta_tracker_single_clip` block records the `anchored_tail_weight` = 1.0 collapse for a single clip and reports `heuristic_variance_across_clips` = 0.0 — an honest reflection of “no across-clip signal on a single-clip render”, not a fabricated statistic.

## Provenance chain

data/gen/provenance\_v1.jsonl — six rows in canonical order (sample\_rules → assemble\_score → xml\_to\_midi → render\_bare → render\_effects → score\_generation), each carrying input\_shas + output\_shas + script + script\_version. The chain reconstructs from any intermediate step forward, and the §21 test asserts stage order and SHA presence.



**Figure 1:** Provenance chain: rule\_id → audio\_sha, five sampled rules through six generation stages to bare + effects renders and the scoring JSON.

## Non-PRNG and non-factor isolation

- PRNG scan of scripts/gen/\*.py: **zero** `^(from|import) (random|numpy\.random|torch|secrets)` matches.
- Non-factor scan: the only textual match on sidecar\_nonfactor under scripts/gen/ is inside assemble\_score.py:46's docstring; the §21 grep guard uses `^(from|import) ...sidecar_nonfactor` at line start and rejects it correctly.
- Interpreter guard assert `sys.executable == "/usr/bin/python3"` present on all six runnable scripts.

## Falsifiability escape hatches

The “byte-determinism failed → hunt source” hatch fired and was resolved honestly: the first-pass scoring JSON leaked full paths and therefore drifted between runs; the fix stored base-names + SHA-256s. All other hatches did not need to fire; the report §9 records which fired and which did not.

## Discussion

Three things about this branch are worth naming.

First, the sampling algorithm's determinism-by-content-hash is the load-bearing invariant. The rules-ledger schema (cycle 6) established content-derived rule\_ids; the rules-ledger extraction (cycle 9) established content-derived event\_ids on ledger rows; this cycle establishes content-derived rule selection. Determinism-by-hash is now the campaign's default answer whenever “how do you pick?” arises, and the answer is legible under audit because every can-

didate's hash is written to the manifest alongside the winner. A future change to the ledger ordering, a future addition of a new candidate rule, or a future edit to any candidate's content will change the sampler's output in a fully-explainable way; a future switch to a PRNG-driven sampler would break the invariant and would be caught by the §21 grep guard.

Second, the rule-composition incoherence is not a bug in the pipeline — it is a real signal from a pipeline built on rules that were selected for content-hash determinism, not for musical coherence. The sampled arrangement rule silences pitched Parts; the sampled harmonic and melodic rules assume pitched Parts exist. The branch publishes this tension rather than patching it away, and the concrete recommendation is a M-GEN-1/rule-composition-constraint sub-milestone in a future cycle: a post-sampling coherence gate that flags when the arrangement silences all pitched Parts, or when form section granularity exceeds the target duration by more than 4×. It must not live in the sampler itself, because that would break the SHA-256 tiebreak's determinism; it lives as a validator on the sampler's output. (One minor documentation refinement noted by the auditor: §8.1's claim that pitched Parts "produce no notes" under the drums-only arrangement is directionally correct but the exported `generated.mid` still contains ~62 note-on events with `velocity > 0` from `music21.harmony.ChordSymbol` writing chord pitches at MIDI export; the pipeline is still non-silent and byte-deterministic, and the directional claim about rule composition stands.)

Third, the uncalibrated-ear caveat is the campaign's biggest open credibility gap. The CORN head is trained on synthetic labels because the rated audio is still egress-blocked, and the head predicts 6/7 on a drum-solo generated from an arbitrary rule composition — a strong signal that the head is uncalibrated. The `ear.calibration = "synthetic_labels_only"` sentinel plumbed through the scoring JSON and re-surfaced in the report is doing its job of preventing this from being read as a musical judgment, and the M-INGEST-1/egress-ready-automation state machine (cycle 8) will fire the retraining pipeline unattended the moment two consecutive fresh `media_ok=true` rows land. Everything downstream — the corn head weights persisted at `data/ear/corn_head_v1.pt`, the feature-version guard, the re-score of this cycle's `effects_layered.wav` for calibrated comparison — is a straight-line consequence of that unblock.

At branch close, M-GEN-1/first-generation is validated/medium and parent M-GEN-1 rolls up to validated/medium. /medium (rather than /high) is the honest grade: the pipeline claim is solid, but the two known limitations (uncalibrated ear, rule-composition incoherence) shave the confidence.

## Open Questions

- **M-GEN-1/rule-composition-constraint** — post-sampling coherence gate that flags arrangement-silences-pitched-Parts or form-granularity-too-fine-for-duration. Runs on the sampler's output, not inside the sampler; preserves the SHA-256 determinism.
- **CORN-head calibration.** When rated audio arrives, retrain the head on real labels, persist weights at `data/ear/corn_head_v1.pt` with a feature-version guard, and re-score this cycle's `effects_layered.wav` so 6/7 can be compared against a calibrated baseline.
- **Longer-duration M-GEN-1 targets.** `duration_s=60` or `90` so more of the sampled form rule's 128-measure structure lands. The current 30 s target was inherited from M-TEX-1/stage-by-stage; longer targets are cheap here.
- **Documentation refinements** (auditor MINOR): update report §11 total-check count from 343 → 413 and §21 count from 39 → 61; tighten §8.1 to note that `ChordSymbol` MIDI realisations still fire even when the arrangement rule leaves pitched Parts otherwise empty. Not blocking.
- **Split the cross-branch integration test** by milestone at ~890 lines. Cheap in scope; correctly deferred to a dedicated future cycle by the worker so it does not entangle a substantive branch.

- **Shadow-ledger adoption at post-merge integration** — 85 orphan-artifact WARNs will clear under `_infra/adopt-fanout-artifacts-m-gen-1` at fork merge.

## Appendix: Provenance

**Cycle range:** cycle 1 of fork `00b3ae64444c`, clone 0. **Working directory:** `/home/user/long-exposure-runs/music-gen`. **Session references:** researcher `c96fe124-a573-407b-9793-4dce6be05ae8`, worker `a9c6730b-5fc1-450f-8e2c-26bdb58cc27d`, auditor `179df4e2-bde2-4aa4-8bc5-6e58b7c16aaa`. **Auditor verdict:** **VALIDATED**. Sub-milestone M-GEN-1/first-generation closes at `validated/medium`; parent M-GEN-1 rolls up at `validated/medium`.

### Deliverables on disk:

- Code: `scripts/gen/{__init__.py, sample_rules.py, assemble_score.py, render_pipeline.py, score_generation.py, emit_provenance.py, plot_gen_report.py}` — interpreter-guarded, zero PRNG imports, no `sidecar_nonfactor` imports.
- Data: `data/gen/{sampling_manifest.json, generated.musicxml, generated.mid, renders/{bare_midi.wav, effects_layered.wav}, scoring_v1.json, provenance_v1.jsonl, render_manifest.json}`.
- Figure: `docs/figures/gen_first_generation_provenance.png`.
- Report: `docs/gen_first_generation_report.md` (397 lines).
- Test: `tests/test_integration_cross_branch.py` §21 — 61 M-GEN-1/first-generation checks including per-artefact SHA-256 anchors, panel-no-aggregate assertion, heuristics-in-[0,1] contract, ear-calibration-sentinel presence, PRNG-import grep guard, and provenance-chain shape.
- Plan of record: line 42 carries the M-GEN-1/first-generation row with all five falsifiable criteria (a)–(e).

**Test state at branch close:** `tests/test_integration_cross_branch.py` — 413 PASS / 0 FAIL, 61 in §21. Live auditor re-run of `scripts.gen.assemble_score` in a scratch directory reproduces the sampling-manifest and MusicXML SHAs byte-identically.

**Environment stack unchanged since cycle 9:** `mscore3 3.2.3 headless`; Python 3.11.15; `numpy 1.26.4`; `music21 9.1.0`; `mir_eval 0.8.2`; `fluidsynth` (Debian) with pinned SF2 `74594e8f...1cb0`; `DawDreamer + Surge XT Effects.vst3` at `/usr/lib/vst3/`; `VGGish` rung on the texture panel; M-EAR-1/preparation CORN head. Single-thread BLAS pins throughout; `DawDreamer` determinism pins applied before import.

**Ledger routing.** Seven M-GEN-1 events written to the per-clone shadow ledger at `/home/user/music-gen-instance/fork-00b3ae64444c/clone-0/promise_ledger.jsonl`. `promise_check .` → 0 ERRORS, 85 orphan-artifact WARNs on `data/gen/*`, `scripts/gen/*`, `docs/gen_first_generation_report.md`, `docs/figures/gen_first_generation_provenance.png` — expected because the root ledger has 156 rows with no M-GEN-1/\* events; adoption happens at fork post-merge integration under `_infra/adopt-fanout-artifacts-m-gen-1`. `org_check .` shows only pre-existing “figure in docs/” WARNs (`docs/figures/gen_first_generation_provenance.png` follows the same location convention as `pipeline_breadth_panel.png` and `tex_stage_by_stage_families.png`) and three root-file WARNs; none introduced by this branch.

**Handoff.** Merge report written to `/home/user/music-gen-instance/fork-00b3ae64444c/clone-0/merge_report.md`. The workspace-root `merge_report.md` is stale (fork `3a908edcb241` clone 2, cycle 8) and should be overwritten at fork rollup with the `00b3ae64444c` rollup once all sibling clones report in.

validated